

SECTION 1. IDENTIFICATION OF THE CHEMICAL AND COMPANY

Product name : AQUANOMIC 2.0 LOW TEMP LAUNDRY SOUR

Other means of identification : Not applicable

Recommended use : Laundry product

Restrictions on use : Reserved for industrial and professional use.

Product dilution information : 0,02 % - 0,1 %

Company : Chile ECOLAB S.A.
Avenida Pedro de Valdivia 3801
Santiago, Santiago Chile
(2)-22413300, Telephone: (2)-22381603, Telephone: SAC: 600 241 6600
sac.chile@ecolab.com

Emergency telephone : (+56-2) 2635-3600 (CITUC) EMERGENCY PHONE

Toxicological Emergency Phone : CITUC (+56-2) 2635-3800 (24 hours) Toxicological Emergencies

SECTION 2. HAZARDS IDENTIFICATION

Classification according to NCH 382 : Class: Class 8 Corrosives

Distintivo según NCh 2190 :


GHS Classification
Product AS SOLD

Skin corrosion : Category 1A

Serious eye damage : Category 1

Product AT USE DILUTION

Not a hazardous substance or mixture.

GHS label elements
Product AS SOLD

Hazard pictograms :



Signal Word : Danger

Hazard Statements : Causes severe skin burns and eye damage.

SAFETY DATA SHEET

AQUANOMIC 2.0 LOW TEMP LAUNDRY SOUR

Precautionary Statements : **Prevention:**
Wash skin thoroughly after handling. Wear protective gloves/ protective clothing/ eye protection/ face protection.
Response:
IF SWALLOWED: Rinse mouth. Do NOT induce vomiting. IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/shower. IF INHALED: Remove person to fresh air and keep comfortable for breathing. Immediately call a POISON CENTER/doctor. IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Immediately call a POISON CENTER/doctor. Wash contaminated clothing before reuse.
Storage:
Store locked up.
Disposal:
Dispose of contents/ container to an approved waste disposal plant.

Product AT USE DILUTION

Precautionary Statements : **Prevention:**
Wash hands thoroughly after handling.
Response:
Get medical advice/ attention if you feel unwell.
Storage:
Store in accordance with local regulations.

Product AS SOLD

Other hazards : Do not mix with bleach or other chlorinated products – will cause chlorine gas.

SECTION 3. COMPOSITION/INFORMATION OF INGREDIENTS

Product AS SOLD

Pure substance/mixture : Mixture

Chemical name	CAS-No.	Concentration (%)
formic acid	64-18-6	10 - 30
citric acid	77-92-9	1 - 5
ammonium chloride	12125-02-9	1 - 5

Product AT USE DILUTION

No hazardous ingredients

SECTION 4. FIRST AID MEASURES

Product AS SOLD

In case of eye contact : Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Get medical attention immediately.

In case of skin contact : Wash off immediately with plenty of water for at least 15 minutes. Use a mild soap if available. Wash clothing before reuse. Thoroughly clean shoes before reuse. Get medical attention immediately.

If swallowed : Rinse mouth with water. Do NOT induce vomiting. Never give anything by mouth to an unconscious person. Get medical attention immediately.

SAFETY DATA SHEET

AQUANOMIC 2.0 LOW TEMP LAUNDRY SOUR

If inhaled	: Remove to fresh air. Treat symptomatically. Get medical attention if symptoms occur.
Protection of first-aiders	: If potential for exposure exists refer to Section 8 for specific personal protective equipment.
Notes to physician	: Treat symptomatically.

Product AT USE DILUTION

In case of eye contact	: Rinse with plenty of water.
In case of skin contact	: Rinse with plenty of water.
If swallowed	: Rinse mouth. Get medical attention if symptoms occur.
If inhaled	: Get medical attention if symptoms occur.

Product AS SOLD

Expected acute effects

Eyes	: Causes serious eye damage.
Skin	: Causes severe skin burns.
Ingestion	: Causes digestive tract burns.
Inhalation	: May cause nose, throat, and lung irritation.

Expected chronic effects

Chronic Exposure	: Health injuries are not known or expected under normal use.
------------------	---

Product AT USE DILUTION

Expected acute effects

Eyes	: Health injuries are not known or expected under normal use.
Skin	: Health injuries are not known or expected under normal use.
Ingestion	: Health injuries are not known or expected under normal use.
Inhalation	: Health injuries are not known or expected under normal use.

Expected chronic effects

Chronic Exposure	: Health injuries are not known or expected under normal use.
------------------	---

Most important symptoms / effects

Product AS SOLD

Eye contact	: Redness, Pain, Corrosion
Skin contact	: Redness, Pain, Corrosion
Ingestion	: Corrosion, Abdominal pain
Inhalation	: Respiratory irritation, Cough

Product AT USE DILUTION

Eye contact	: No symptoms known or expected.
Skin contact	: No symptoms known or expected.
Ingestion	: No symptoms known or expected.

SAFETY DATA SHEET

AQUANOMIC 2.0 LOW TEMP LAUNDRY SOUR

Inhalation : No symptoms known or expected.

SECTION 5. FIREFIGHTING MEASURES

Product AS SOLD

- Extinguishing Agents : Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.
- Unsuitable extinguishing agents : None known.
- Specific hazards during fire fighting : Not flammable or combustible.
- Products generated in combustion and thermal degradation : Decomposition products may include the following materials:
Carbon oxides
Nitrogen oxides (NO_x)
Sulfur oxides
Oxides of phosphorus
- Precautions for Emergency Personnel and / or Firefighters : Use personal protective equipment.
- Specific extinguishing methods : Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations. In the event of fire and/or explosion do not breathe fumes.

SECTION 6. MEASURES IN CASE OF ACCIDENTAL SPILLAGE

Product AS SOLD

- Personal precautions, protective equipment and emergency procedures : Ensure adequate ventilation. Keep people away from and upwind of spill/leak. Avoid inhalation, ingestion and contact with skin and eyes. When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Ensure clean-up is conducted by trained personnel only. Refer to protective measures listed.
- Environmental precautions : Do not allow contact with soil, surface or ground water.
- Methods and materials for containment and cleaning up : Stop leak if safe to do so. Contain spillage, and then collect with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13). Flush away traces with water. For large spills, dike spilled material or otherwise contain material to ensure runoff does not reach a waterway.

Product AT USE DILUTION

- Personal precautions, protective equipment and emergency procedures : Refer to protective measures listed.
- Environmental precautions : No special environmental precautions required.
- Methods and materials for containment and cleaning up : Stop leak if safe to do so. Contain spillage, and then collect with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to

SAFETY DATA SHEET

AQUANOMIC 2.0 LOW TEMP LAUNDRY SOUR

local / national regulations (see section 13). Flush away traces with water. For large spills, dike spilled material or otherwise contain material to ensure runoff does not reach a waterway.

SECTION 7. HANDLING AND STORAGE

Product AS SOLD

Handling

Advice on safe handling : Do not ingest. Do not get in eyes, on skin, or on clothing. Do not breathe dust/ fume/ gas/ mist/ vapors/ spray. Use only with adequate ventilation. Wash hands thoroughly after handling. Do not mix with bleach or other chlorinated products – will cause chlorine gas.

Storage

Conditions for safe storage : Keep away from strong bases. Keep out of reach of children. Keep container tightly closed. Store in suitable labeled containers.

Recommended and unsuitable packaging by supplier : Use high density plastic containers.

Storage temperature : 0 °C to 50 °C

Product AT USE DILUTION

Handling

Advice on safe handling : Wash hands after handling. For personal protection see section 8.

Storage

Conditions for safe storage : Keep out of reach of children. Store in suitable labeled containers.

Recommended and unsuitable packaging by supplier : Use high density plastic containers.

SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Product AS SOLD

Ingredients with workplace control parameters

Components	CAS-No.	Form of exposure	Permissible concentration	Basis
formic acid	64-18-6	CMP	5 ppm	AR OEL
		CMP - CPT	10 ppm	AR OEL
formic acid	64-18-6	LT	4 ppm 7 mg/m ³	BR OEL
formic acid	64-18-6	LPP	4,4 ppm 8,2 mg/m ³	CL OEL
		LPT	10 ppm 19 mg/m ³	CL OEL
formic acid	64-18-6	TWA	5 ppm 9,4 mg/m ³	PE OEL
		STEL	10 ppm 18,8 mg/m ³	PE OEL
formic acid	64-18-6	L-8/40	5 mg/m ³	VE OEL
formic acid	64-18-6	VLE-PPT	5 ppm	NOM-010-STPS-2014
		VLE-CT	10 ppm	NOM-010-STPS-2014

SAFETY DATA SHEET

AQUANOMIC 2.0 LOW TEMP LAUNDRY SOUR

formic acid	64-18-6	TWA	5 ppm	ACGIH
		STEL	10 ppm	ACGIH
		TWA	5 ppm 9 mg/m3	NIOSH REL
		TWA	5 ppm 9 mg/m3	OSHA Z1
ammonium chloride	12125-02-9	CMP (Fumes)	10 mg/m3	AR OEL
		CMP - CPT (Fumes)	20 mg/m3	AR OEL
ammonium chloride	12125-02-9	TWA (Fumes)	10 mg/m3	PE OEL
		STEL (Fumes)	20 mg/m3	PE OEL
ammonium chloride	12125-02-9	L-8/40 (Fumes)	10 mg/m3	VE OEL
		LB (Fumes)	20 mg/m3	VE OEL
ammonium chloride	12125-02-9	VLE-PPT (Fumes)	10 mg/m3	NOM-010-STPS-2014
		VLE-CT (Fumes)	20 mg/m3	NOM-010-STPS-2014
ammonium chloride	12125-02-9	TWA (Fumes)	10 mg/m3	ACGIH
		STEL (Fumes)	20 mg/m3	ACGIH
		TWA (Fumes)	10 mg/m3	NIOSH REL
		STEL (Fumes)	20 mg/m3	NIOSH REL

Engineering measures : Effective exhaust ventilation system. Maintain air concentrations below occupational exposure standards.

Personal protective equipment

Eye protection : Safety goggles
Face-shield

Hand protection : Wear the following personal protective equipment:
Impervious gloves, resistant to chemicals.
Gloves should be discarded and replaced if there is any indication of degradation or chemical breakthrough.

Skin protection : Personal protective equipment comprising: suitable protective gloves, safety goggles and protective clothing

Wear closed shoes.

Respiratory protection : When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.

Hygiene measures : Handle in accordance with good industrial hygiene and safety practice. Remove and wash contaminated clothing before re-use. Wash face, hands and any exposed skin thoroughly after handling. Provide suitable facilities for quick drenching or flushing of the eyes and body in case of contact or splash hazard.

Product AT USE DILUTION

Engineering measures : Good general ventilation should be sufficient to control worker exposure to airborne contaminants.

Personal protective equipment

Eye protection : No special protective equipment required.

SAFETY DATA SHEET

AQUANOMIC 2.0 LOW TEMP LAUNDRY SOUR

Hand protection	: No special protective equipment required.
Skin protection	: No special protective equipment required.
Respiratory protection	: No personal respiratory protective equipment normally required.

SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

	Product AS SOLD	Product AT USE DILUTION
Appearance	: liquid	liquid
Color	: clear, light blue	colorless
Odor	: pungent	not significant
pH	: 0,7 - 1,0, (100 %)	4,5 - 7,5
Flash point	: Not applicable, Does not sustain combustion.	
Odor Threshold	: No data available	
Melting point/freezing point	: No data available	
Initial boiling point and boiling range	: > 100 °C	
Evaporation rate	: No data available	
Flammability (solid, gas)	: No data available	
Upper explosion limit	: No data available	
Lower explosion limit	: No data available	
Vapor pressure	: No data available	
Relative vapor density	: No data available	
Relative density	: 1,036 - 1,056	
Water solubility	: soluble	
Solubility in other solvents	: No data available	
Partition coefficient: n-octanol/water	: No data available	
Autoignition temperature	: No data available	
Thermal decomposition	: No data available	
Viscosity, kinematic	: No data available	
Explosive properties	: No data available	
Oxidizing properties	: The substance or mixture is not classified as oxidizing.	
Molecular weight	: No data available	
VOC	: No data available	

SECTION 10. STABILITY AND REACTIVITY

Product AS SOLD

Chemical stability	: Stable under normal conditions.
Possibility of hazardous reactions	: Do not mix with bleach or other chlorinated products – will cause chlorine gas.
Conditions to avoid	: None known.

SAFETY DATA SHEET

AQUANOMIC 2.0 LOW TEMP LAUNDRY SOUR

Incompatible materials	: Chlorine-based bleaching agents Bases Metals
Hazardous decomposition products	: In case of fire hazardous decomposition products may be produced such as: Carbon oxides Nitrogen oxides (NOx) Sulfur oxides Oxides of phosphorus

SECTION 11. TOXICOLOGICAL INFORMATION

Information on likely routes of exposure : Inhalation, Eye contact, Skin contact

Toxicity

Product AS SOLD Product

Acute oral toxicity	: Acute toxicity estimate : > 5.000 mg/kg
Acute inhalation toxicity	: No data available
Acute dermal toxicity	: No data available
Skin corrosion/irritation	: No data available
Serious eye damage/eye irritation	: No data available
Respiratory or skin sensitization	: No data available
Carcinogenicity	: No data available
Reproductive effects	: No data available
Germ cell mutagenicity	: No data available
Teratogenicity	: No data available
STOT-single exposure	: No data available
STOT-repeated exposure	: No data available
Aspiration toxicity	: No data available

SECTION 12. ECOLOGICAL INFORMATION

Product AS SOLD Ecotoxicity

Environmental Effects : This product has no known ecotoxicological effects.

Product

Toxicity to fish	: No data available
Toxicity to daphnia and other aquatic invertebrates	: No data available
Toxicity to algae	: No data available

Components

SAFETY DATA SHEET

AQUANOMIC 2.0 LOW TEMP LAUNDRY SOUR

Toxicity to fish : formic acid
96 h LC50: > 100 mg/l

citric acid
96 h LC50 Fish: > 100 mg/l

Persistence and degradability

Product AS SOLD

Readily biodegradable.

Product AT USE DILUTION

Readily biodegradable.

Bioaccumulative potential

No data available

Mobility in soil

No data available

Other adverse effects

No data available

SECTION 13. INFORMATION OF DISPOSAL

Product AS SOLD

Disposal methods : Where possible recycling is preferred to disposal or incineration. If recycling is not practicable, dispose of in compliance with local regulations. Dispose of wastes in an approved waste disposal facility.

Disposal considerations : Dispose of as unused product. Empty containers should be taken to an approved waste handling site for recycling or disposal. Do not re-use empty containers. Dispose of in accordance with local, state, and federal regulations.

Disposal of contaminated packaging : In accordance with what is described by Supreme Decree No. 148, the packaging of the product is considered hazardous waste and must be disposed of through companies authorized to receive and / or treat such waste, which must be issued and final disposal certificate waste.

Product AT USE DILUTION

Disposal methods : Diluted product can be flushed to sanitary sewer.

Disposal considerations : Dispose of in accordance with local, state, and federal regulations.

Disposal of contaminated packaging : In accordance with what is described by Supreme Decree No. 148, the packaging of the product is considered hazardous waste and must be disposed of through companies authorized to receive and / or treat such waste, which must be issued and final disposal certificate waste.

SECTION 14. TRANSPORT INFORMATION

Product AS SOLD

The shipper/consignor/sender is responsible to ensure that the packaging, labeling, and markings are in compliance with the selected mode of transport.

Land transport

SAFETY DATA SHEET

AQUANOMIC 2.0 LOW TEMP LAUNDRY SOUR

Official transport classification of the United Nations

UN number	:	3412
Description of the goods	:	FORMIC ACID (II)
Class	:	8
Packing group	:	II
Environmentally hazardous	:	no

Air transport (IATA)

Official transport classification of the United Nations

Contact Regulatory for air freight eligibility

Sea transport (IMDG/IMO)

Official transport classification of the United Nations

UN number	:	3412
Description of the goods	:	FORMIC ACID (II)
Class	:	8
Packing group	:	II
Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code	:	Not applicable
Marine pollutant	:	no

SECTION 15. REGULATORY INFORMATION

Registrations and Certifications

Chile: Our HDS meets Chilean Normative: NCH 2245: 2015 (Safety Data Sheet for chemical products - Content and order of sections)

The receiver should pay attention to the possible existence of local regulations.

NCh 1411: Prevention of risks, IV identification of risks of materials

NCh 2190: Transportation of Hazardous Substances - Pictograms for hazard identification

NCh 382: Dangerous Good - Classification

D.S. No. 594: Minimum basic conditions in the workplace

D.S. No. 148: Hazardous waste disposal

D.S. No. 72: Mining Safety Regulations

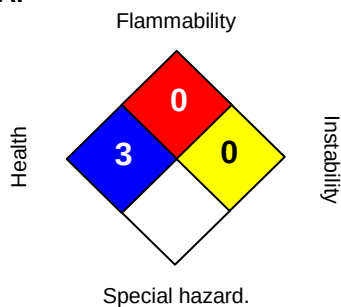
D. S. No. 43: reports on the storage of hazardous substances

D. S. No. 40: Reporting on exposure hazards

SECTION 16. OTHER INFORMATION

Product AS SOLD

NFPA:



HMIS III:

HEALTH	3
FLAMMABILITY	0
PHYSICAL HAZARD	0

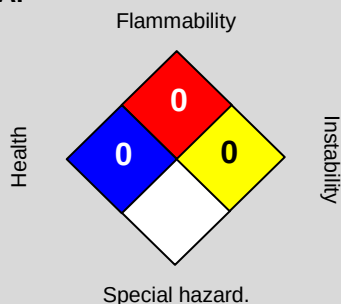
0 = not significant, 1 = Slight,
2 = Moderate, 3 = High
4 = Extreme, * = Chronic

Product AT USE DILUTION

SAFETY DATA SHEET

AQUANOMIC 2.0 LOW TEMP LAUNDRY SOUR

NFPA:



HMIS III:

HEALTH	0
FLAMMABILITY	0
PHYSICAL HAZARD	0

0 = not significant, 1 = Slight,
2 = Moderate, 3 = High
4 = Extreme, * = Chronic

Issuing date : 09.11.2018
Version : 1.1
Prepared by : Regulatory Affairs

While there is no change in the formula or hazard classifications, this SDS remains valid.

REVISED INFORMATION: Significant changes to regulatory or health information for this revision is indicated by a bar in the left-hand margin of the SDS.

The information provided in this Material Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.